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Design Report: Design, Analysis and Manufacturing of Front Suspension and Steering System for an All-Terrain Vehicle

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ABSTRACT: This article goes into the complete process of designing and manufacturing the suspension and steering system of an All-terrain vehicle that competed at the Baja SAEINDIA competition. The kinematic analysis of the linkages was performed in LOTUS Shark software to arrive at an optimal geometry for the front suspension and steering system. Preliminary designs were modelled in SOLIDWORKS and were further optimized after performing Finite Element Analysis (FEA) simulations in ANSYS Workbench. A Short-Long Double Wishbone setup was selected for the front suspension. The steering geometry was designed according to the Ackermann principle. The article also discusses the design and manufacturing of individual components of these systems to finally have a setup that is lightweight and cost-effective.

Keywords: All-Terrain Vehicle, Double wishbone suspension, Knuckle, Ackerman steering, Rack and Pinion gearbox.

Abbreviations: SAE, Society of Automotive Engineers, CNC, Computer Numerical Control

I. INTRODUCTION

Baja SAEINDIA is an intercollegiate engineering competition consisting of various static and dynamic events in which students function as a team to design, fabricate, test and compete with an All-terrain vehicle. All the systems of the vehicle need to comply with the rulebook of the competition which sets the design limitations. The suspension and steering system of an All-terrain vehicle is designed to get the optimum performance in acceleration, handling, braking and to also provide comfort to the driver on rough terrains during the ride. Furthermore, the design needs to be lightweight as well as reliable to maximize the performance and at the same time endure severe loads. The vehicle's suspension was to be designed keeping in mind the harsh terrain it must endure. The suspension system must be rugged enough to absorb the shocks from bumps, drops, logs and rocks as well as provide good

control of the car with a nice ride experience. Hence a 4-wheel independent suspension system was chosen which provides optimal performance to the vehicle [1]. The front suspension geometry was selected as a short-long arm. This suspension setup allows to have good adjustability over various parameters such as camber, caster, roll center position. It also has camber gain during cornering that facilitates good road contact during tight turns [2]. The steering system was designed to effectively maneuver the vehicle over rough terrain and give precise control during cornering. A manual rack and pinion mechanism was selected as it gives less backlash and a compact design.

II. METHODOLOGY

A. DESIGN CONSIDERATIONS/PARAMETERS

Camber: Camber is defined as the angle between a tilted wheel plane and the vertical when viewed from the front.

Camber is considered positive if the top of the wheel leans outwards and negative if the top of the wheel leans inwards. Both positive and negative camber provides uneven tire wear. But while negotiating a curve the outer wheel will experience a positive camber due to lateral force that can be compensated by providing a slight negative camber (close to zero). Negative camber gain on inner wheels and positive camber gains on the outer wheel will ease the cornering [3].

Caster angle: Caster angle can be defined as the angle made by the kingpin in the side view with the vertical axis passing through the wheel center. Providing positive caster allows the self-centering of the steering wheels. Providing a negative caster will harden the steering.

Toe: Toe is the symmetric angle that each wheel makes with the longitudinal axis of the vehicle. Tire wear is heavily dependent on toe distances. Dynamic factors induce a change in toe which leads to two conditions, toe-in and toe-out.

Kingpin angle: Kingpin inclination is defined as the angle at which the kingpin axis is inclined to the vertical axis passing through the wheel center. Kingpin angle affects the performance of the car when the wheels are steered. More the kingpin angle more the car will lift when steered. When the kingpin axis is extended up to the ground, it intersects the ground at a particular point. The distance of that point from the center of the wheel contact patch is called the scrub radius. Scrub radius increases the wear of the tire but some amount of negative scrub radius is required so that the wheel purely rolls when steered [4].

Roll center: The Roll Center of a suspension system is that point in the transverse plane of the axles, about which the sprung mass of that end of the vehicle will roll under the influence of that end of the vehicle will roll under the influence of centrifugal force [4].

Ackermann Geometry: The steering geometry of the linkages was designed according to the Ackermann principle where the front inside tire turns more than the front outside tire during a turn [4]. This geometry was selected as the vehicle is not cornering at very high speeds where there is a large amount of load transfer on the outside tires. Therefore, there is no significant difference between the slip angles of the inside and outside tires. In an ideal case of Ackermann geometry, all the 4 tires will be in pure rolling condition about an instantaneous center of rotation. The ratio of the difference in steering angles for the given geometry to the difference in steering angles for the ideal case multiplied by 100 gives the Ackermann percentage. The value of Ackermann percentage indicates how close the given geometry is to the ideal case.

B. GEOMETRIC ANALYSIS

Based on the preliminary front suspension geometry, wishbone design and limitations given by the Baja SAEINDIA rulebook, the hardpoints on the chassis were determined. The ideal travel for an all-terrain vehicle with the sprung mass considerations was evaluated to be 4" in bump and 2" in droop. The static camber of the wheels was taken as $+3^\circ$, to ensure the stability of the vehicle in dynamic conditions. These values were given as an input to the LOTUS Shark software. The graph in Fig. 1 gives the camber angle variation of the front wheels as they travel upwards(bump) and then downwards(droop). The camber is positive in droop with a maximum of $+2^\circ$ and becomes negative in the bump condition with a maximum of -8° ; this will increase the traction of the wheel. The final suspension parameters are shown in Table 1. And the static suspension geometry in Fig. 2.

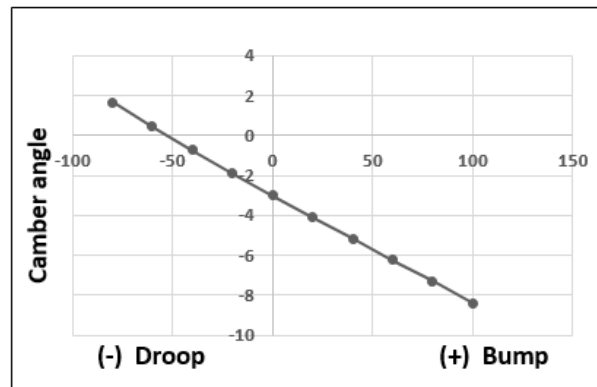


Fig. 1. Camber change with wheel travel

Table 1. Suspension Parameters

Parameters	Values
Static camber	-3°
Camber change	$+2^\circ$ to -8°
Caster angle	$+8^\circ$
Kingpin inclination angle	$+8^\circ$
Toe-in	2°
Scrub radius	2.6 inches
Ground clearance	12 inches
Roll center height	10.24 inches

The geometric analysis of the steering system was also done on LOTUS SHARK software. The aim while performing the analysis was to maximize Ackermann percentage and minimize both the turning radius and bump steer. Front steer configuration is being used in which the tie rod linkage connects to the steering arm in front of the wheel center. This makes enough space in the cockpit to accommodate the pedals.

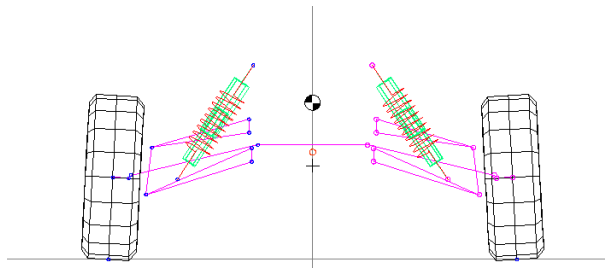


Fig. 2. Static suspension geometry in LOTUS

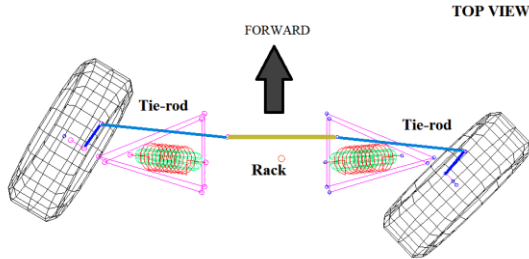


Fig. 3. Limiting steering angles in LOTUS

Many iterations were carried out to finally arrive at an optimum geometry. This was done by precisely changing the location of the inner and outer tie rod endpoints. 80% of Ackermann was obtained at maximum rack travel of 58 mm from the center to lock. The limiting steering angles were 50° and 32° for the front inside and outside tires respectively as shown in Fig. 3. The turning radius at these angles is 1.8 m. Other parameters such as length and position of the rack, length of the tie rod and the length of the steering arm were obtained from this geometry. The variation of Ackermann % with respect to rack travel is shown in Fig. 4.

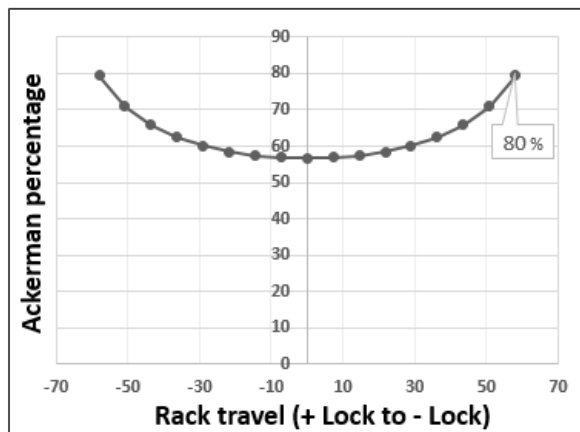


Fig. 4. Ackermann % with rack travel

Bump steer was also minimized as shown in Fig. 5 by analyzing the dynamic toe in bump and droop.

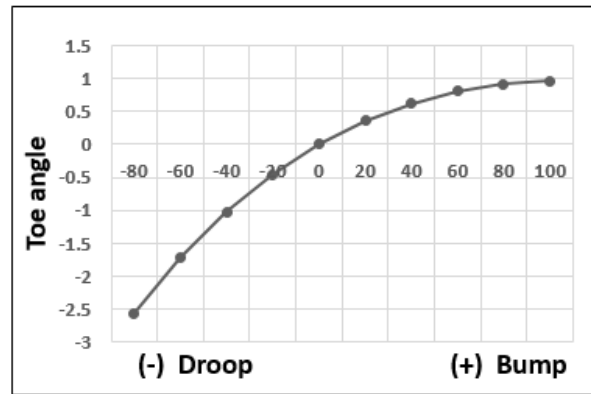


Fig. 5. Change in toe angle with wheel travel

The radius of the pinion plays an important role in the design of the steering system. Increasing the radius decreases the steering wheel revolutions from lock to lock but increases the steering effort which is not desirable as a smaller pinion gives a better mechanical advantage than a larger one. This interdependency of the quantities was taken into consideration when arriving at the final setup shown in Table 2.

Table 2. Rack and Pinion parameters

Rack length	460 mm
Rack travel lock to lock	58 mm
Diameter of pinion	46 mm
Module	2
Number of teeth on pinon	23
Pressure angle	20°

C. STEERING RATIO

It is defined as the ratio of steering wheel angle (in degrees) to the road wheel angle (in degrees) from lock to lock [4]. But, as this setup is related to Ackermann geometry the steering ratio is nonlinear, that is, the ratio varies with wheel angle [4]. A graph is plotted between road wheel angles and steering wheel angle using data points obtained from LOTUS shark software as shown in Fig.6. The average slope of these data points gives the overall steering ratio as 3.57.

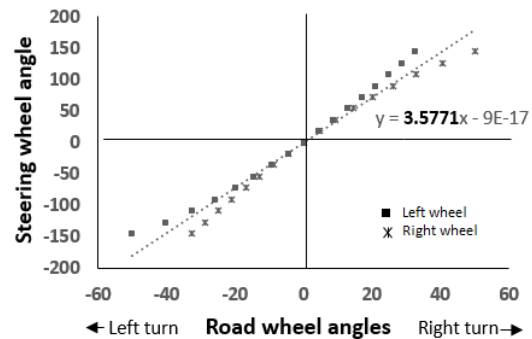


Fig. 6. Steering ratio

D. STEERING EFFORT

Steering effort is defined as the torque needed at the steering wheel to turn the road wheels against various resistances. The resistive torque produced by the static frictional force at the steering axis when the vehicle is stationary accounts by far the largest of all [5].

Mass of vehicle with driver = 250 Kg

For 33% weight bias on front axle the load on each front wheel is $(250 \times g \times 0.33) / 2 = 404.66 \text{ N}$

Friction force at the contact patch assuming $\mu=0.7$

$$= 0.7 \times 404.66 = 283.26 \text{ N}$$

This force which is at a perpendicular distance of 70mm from steering axis creates a resistive torque about it

$$= 283.26 \text{ N} \times 0.07 = 19.82 \text{ N m}$$

This torque should be overcome by the torque produced by the force acting at the steering arm

The perpendicular force required at the steering arm is equal to the resistive torque divided by the length of steering arm (length of steering arm = 102 mm)

$$= (19.82) / (0.102) = 194.4 \text{ N}$$

This is the component of the force that should be applied by the tie rod since it is at an angle of 20° with the horizontal plane as illustrated in Fig. 7. The tie rod is perpendicular to the steering arm when viewed from the top. Therefore, force applied by tie rod is

$$= 194.4 / (\cos(20)) = 206.87 \text{ N}$$



Fig. 7. Tie rod angle

The force that should be applied by the rack as it is also at an angle of 20° to the tie rod is

$$= 206.87 / (\cos(20)) = 220 \text{ N}$$

The rack is pushing one tie rod with this force and pulling the other with the same force.

Therefore, the total linear force along the rack is 440 N

The torque needed at the pinion is force along the rack multiplied by the radius of the pinion.

$$440 \times 0.023$$

$$= 10.1 \text{ N-m}$$

Torque applied at the pinion is the same as the torque applied at the steering wheel. The magnitude of force required on the steering wheel is found by dividing this torque by the radius of the steering wheel which is 140 mm

$$= 10.1 / 0.14 = 72.1 \text{ N}$$

This is the force required when the vehicle is stationary and its value further goes down when the vehicle is moving.

E. DESIGN AND MANUFACTURING

1. KNUCKLE

In automotive suspension, the steering knuckle is the component to which the wheel is mounted via the hub which is attached to the knuckle through the spindle on a bearing. The knuckle is connected to the chassis through the wishbones or A-arms. The knuckle helps in turning the wheels through the steering arm which is connected to the tie rod [6]. The knuckle was designed on SOLIDWORKS. The material selected was EN 8 carbon steel due to its low cost and high strength. The various loads acting on the knuckle are shown in Fig. 8

Bump Load: When the vehicle undergoes a ground disturbance in the form of a bump or a hole a force is exerted on the knuckle. For design considerations, it is assumed that the suspension spring has attained full travel which is 4 inches after a jump height of 1.83m. The vehicle mass with driver is calculated as 250 kg from the CAD model.

$$\text{Velocity at impact} = v^2 = 2 \times g \times h; v = 6 \text{ m/s}$$

$$\text{Force on all 4 wheels during impact} = mv / t$$

$$= 250 \times 6 / 0.10 = 15000 \text{ N}; t = \text{time of impact}$$

If vehicle lands on 2 wheels, force on single wheel = 7500N hence, Force on the Assembly = 7500N ~ 3G.

Here G = mass of the vehicle with the driver $\times 9.81 \text{ m/s}^2$

Cornering Load: Lateral force or side force is the cornering force produced by a vehicle on the tire during cornering. It is equivalent to the centrifugal force generated due to cornering. It is determined by the position of the center of gravity and thus the amount of load and cornering force carried by the tire is proportional to the distance of the center of gravity from the tire [7]. The cornering force was calculated and a 2G cornering force was applied to the knuckle.

Lateral Load Transfer during braking: During braking, there is a lateral load transfer from the rear wheels to the front wheels due to the lateral acceleration imposed on the center of gravity (CG) of the car. The force was calculated and a 2G braking load transfer force was applied on the knuckle.

Steering Load: This is the force exerted by the steering gearbox on the steering arm mounting of the knuckle through the tie rod while turning. A 1000N steering force was applied on the steering arm.

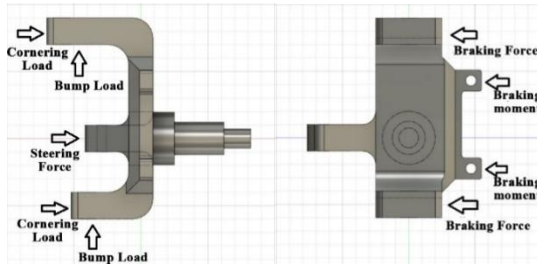


Fig. 8. Forces acting on the knuckle

From the analysis results, the knuckle had a factor of safety of 1.6 and had a max deformation of 0.56mm. After the analysis, the design was finalized in the CAD model and was sent for manufacturing. Equivalent stress distribution is shown in Fig. 9.

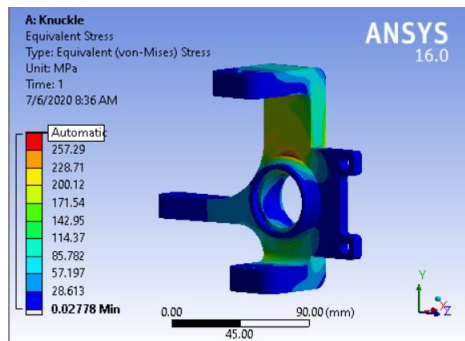


Fig. 9. Equivalent (Von-Mises) Stress in the knuckle

Manufacturing of the knuckle was done on CNC milling. Fig. 10 shows both the manufactured knuckles. The stub axle was manufactured separately on a conventional lathe and was press fit in the knuckle to save material and machining costs.



Fig. 10. Manufactured Knuckles

2. HUB

The hub was designed on SOLIDWORKS and analysis was done on ANSYS Workbench. The forces applied on the hub were a 3G force in bump, 2G cornering force and 2G braking force. From the analysis results, the hub had a factor of safety of 2.3 and max deformation is 0.02 mm. Equivalent stress distribution is shown in Fig. 11.

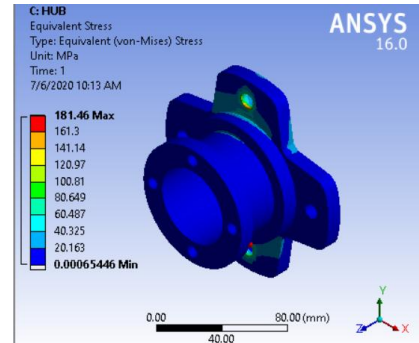


Fig. 11. Equivalent (Von-Mises) Stress in the hub

The hub was designed as 2 separate pieces that were to be welded together to save material and machining costs. The hub was manufactured on the conventional lathe and then tig welded as shown in Fig. 12. The material selected for the hub was also EN 8 carbon steel due to its ease of availability, low cost and high strength.



Fig. 12. Manufactured Hub

3. BEARING SELECTION

The bearing is an important component which connects the hub and the knuckle. It needs to handle the radial and axial forces that are transmitted to the knuckle from the hub. So, taking this into consideration OEM bearings were selected. The bearing we selected was a double row angular contact ball bearing of Mahindra Logan.

4. CONTROL ARMS

A double-wishbone suspension system of unequal length and parallel was implemented for Front suspension. The advantage of this setup is that a negative camber is induced when the suspension goes in bump which provides stability to the car. Negative camber is important to maintain ground contact.

The lengths of the A-arms were obtained from the LOTUS Shark software as 17.4 inches of the lower A-arm and 15.8 inches of the upper A-arm. The spring-damper arrangement is mounted on the lower wishbone with a motion ratio of 0.63 and the wishbone is attached to the knuckle through a ball joint. The positive static camber provided by the wishbone arrangements provides a reduction of the scrub radius of the wheel-steering knuckle combination. This reduces the steering effort required and the rate of tire wear. All of these increase the comfort of the driver which is one of the most important functions of a suspension system.

The static analysis was done on the A-arms on ANSYS Workbench as shown in Fig. 13 by fixing the pivot points and applying the load on the spring mount point. This was to replicate the worst-case scenario when the vehicle had taken a bump and the shock travel was maxed out. The A-arms were made from tubular AISI 4130 steel. The upper A-arm had the dimensions of $\varnothing 25.4 \times 1$ mm wall thickness and Lower A-arm had the dimension of $\varnothing 29.2 \times 1.65$ mm wall thickness. The shock was mounted on the lower a-arm.

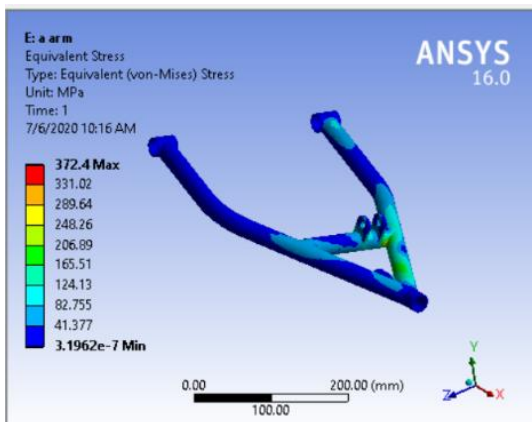


Fig. 13. Equivalent (Von-Mises) Stress in A-Arm

5. SELECTION OF SHOCK ABSORBERS

For the front shock absorbers, FOX FLOAT 3 EVOL R were selected (Fig. 14). These shock absorbers use air as springs, instead of heavy steel coil springs which makes them lightweight and gives a progressive spring rate. These air shock dampers contain high-pressure nitrogen gas and FOX high viscosity index shock oil separated by an Internal Floating Piston system. This helps to ensure consistent, fade-free damping in most riding conditions. These shocks are built using 6061-T6 aluminum for lightweight and strength.

The shock's travel is 5.3 inches and a motion ratio of 0.63 is used to mount the shocks. The air pressure in the shocks is set at 150/50 psi for the EVOL and main chamber respectively which gives a progressive spring

rate from 35 N/mm to 65 N/mm. The complete wheel assembly is shown in Fig. 15.



Fig. 14. Fox Float 3 Evol R air suspension

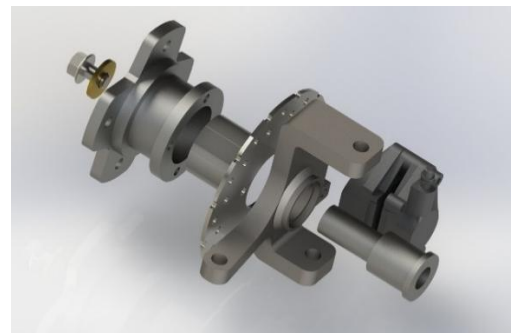


Fig. 15. Exploded view of wheel assembly

6. RACK AND PINION GEARS:

The pitch circle diameter of the pinion is 46 mm as this gives a total of 288° (0.8 revolutions) of steering wheel rotation from lock to lock. Its face width is 20 mm and has stepped shafts on both sides to press-fit the bearings. Static structural analysis of the pinion was done by applying a load of 3000 N on the face of its tooth. This results in a minimum FOS of 1.9 at the tooth root fillet as shown in Fig. 16.

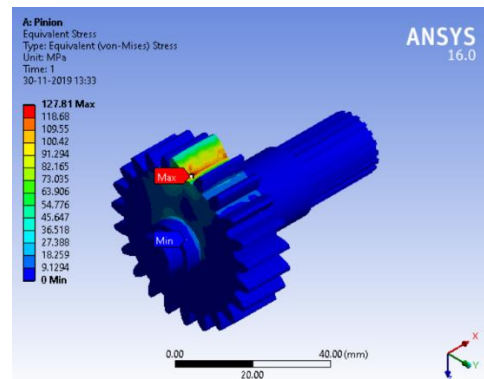


Fig. 16. Equivalent (von-Mises) Stress in Pinion

The material selected for the gears was EN-353 steel due to its availability, low cost and as it provides properties

that are acceptable for several automobile applications. The gears were machined on a gear hobbing machine (Fig. 17) and after which they were subjected to case hardening to increase the wear strength.



Fig. 17. Manufactured Rack and Pinion gears

7. CASING

A casing was designed to support bearings and house the rack and pinion gears. The non-axial forces taken by the rack are counteracted by the casing. Static structural analysis was performed as shown in Fig. 18 by applying a load of 9000 N and adding fixed supports at the mounting points. Minimum FOS obtained was 8.6.

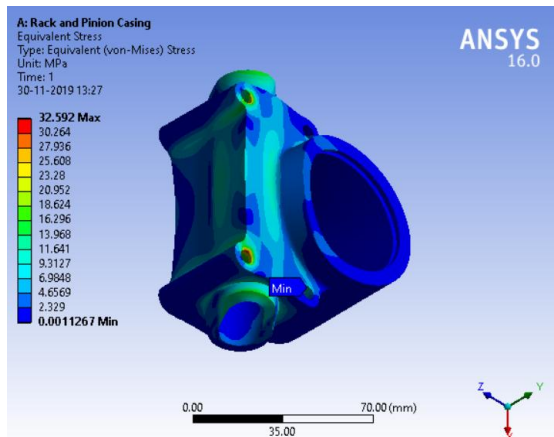


Fig. 18. Equivalent (von-Mises) Stress distribution

The material selected was Aluminum 6061-T6 due to its lightweight and good yield strength. The casing was manufactured on a CNC mill as shown in Fig. 19. (a). Brass bushes were inserted where the rack slides. They help in preventing wear of the casing's inner surface. An internal groove was made to place the retaining ring that holds components in the casing. The casing is mounted on the roll cage with the help of bolts.



Fig. 19. (a) CNC milling operation (b) Finished Casing

8. CLEVIS:

Custom clevises were designed to provide the correct location for the tie rod to mount to the rack. This helped in the prevention of bump steer. They were connected to the rack with socket head screws and locked in position by set screws. Structural analysis (Fig. 20) was done by applying a force of 9400 N at an angle of 45°. It results in a minimum FOS of 1.5.

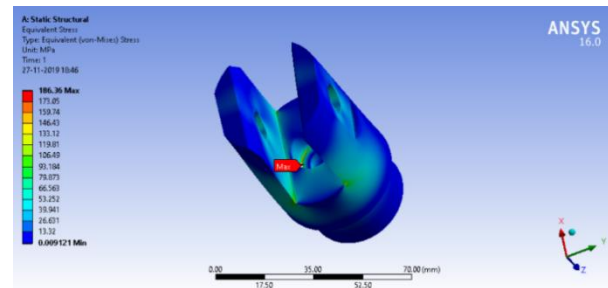


Fig. 20. Equivalent (Von-Mises) Stress in clevis

The material selected was Aluminum 6061-T6. Lathe and mill operations (Fig. 21. (a)) were carried out to manufacture the desired component. Grooves were also made for attaching rubber boots that seal in the lubricant and keep out dirt and water while still allowing movement of the rack.

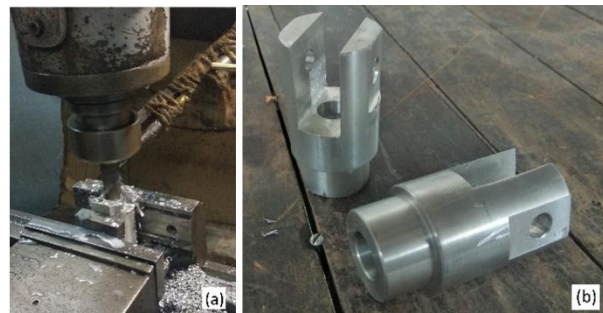


Fig. 21. (a) Milling operation (b) Manufactured Clevises

9. TIE RODS:

The length of the tie rod obtained from LOTUS was 330 mm between pivot points. Tie rod was connected to the clevis via heim joint and the other end was connected to the steering arm with a ball joint. The material selected was Aluminum 6061 T-6 round bar of 20 mm diameter. Heim joint end of the tie rod is left hand threaded and ball joint end is right hand threaded. This allows the tie rod to be rotated after installation for easy adjustment of static toe angle. Buckling analysis calculations give the critical buckling load as 3044 N.

10. COLUMN:

Two pinch bolt universal joints were used in the column to connect the pinion to the steering wheel. This was due to driver preference in the position of the steering wheel. Both the u-joints were angled at 25°. The lower column was a 135 mm long EN 8 round bar which was welded to both u-joints. The upper column was made of Aluminum 6061 T-6. It was 270 mm long and had a flange welded at the end to which the steering wheel was fastened. The added advantage of using two u- joints was that the fluctuations in angular velocity due to both cancel each other.

11. STEERING WHEEL:

The base of the steering wheel was a water-jet cut out of an aluminum 6061 T-6 plate of 5 mm thickness to provide the necessary structural rigidity. The structural analysis was done in Ansys Workbench as shown in Fig. 22 by applying 400 N-m of torque on the wheel. Minimum FOS is 5.6.

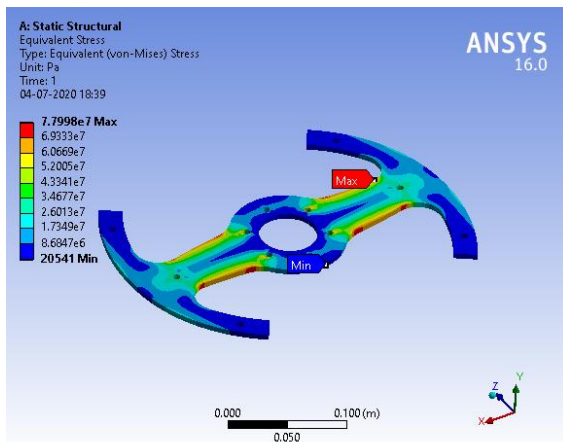


Fig. 22. Equivalent (Von-Mises) Stress in the wheel

Custom 3D printed ergonomic grips of Polylactic acid (PLA) were fastened to the wheel as shown in Fig. 23. The complete assembly of the steering subsystem is shown in Fig. 24.

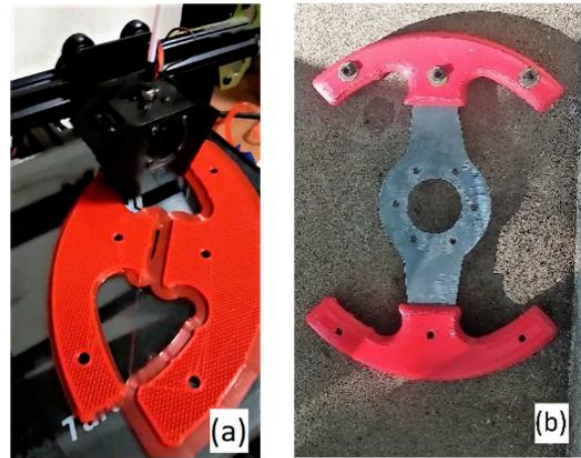


Fig. 23. (a) 3D-printing of grips (b) Steering wheel



Fig. 24. Steering subsystem assembly

IV. CONCLUSION

The goal of designing and manufacturing suspension and steering systems that were cost-effective and reliable was accomplished. The geometric analysis of the system was done on LOTUS shark and an optimum geometry was achieved. The modelling was done on SOLIDWORKS and FEA simulation was performed on ANSYS Workbench for different materials which showed the components had a good factor of safety. The components were successfully manufactured and assembled on the vehicle These sub-systems were used in the ATV of Muffakham Jah College of Engineering and Technology at Baja SAEINDIA 2020. The reliability of the system was proven in the endurance race where the vehicle ran for 4 hours on off-road conditions without breaking down.

V. FUTURE SCOPE

In the suspension system, the weight of the knuckle can be reduced by using Aluminum 7075-T6 material. Deep groove single row ball bearings can be used in place of the OEM bearing to further reduce weight. The weight of the steering system can be further reduced by opting for a setup that does not require universal joints.

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Conflict of Interest: Nil

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